Practice Problem 2 Guide Questions

In case the student's code does not compile properly OR are getting warnings due to sqrt() being undefined despite having #include <math.h>, it is likely that their gcc compilation is missing the -lm flag which requires the linker to provide the definition of the sqrt() function. The correct gcc command is: gcc practice2.c -o practice2 -lm

1. HID
2. BIG: 681. DAY cannot be generated because Y would be represented by 24, which is outside of our single digit range.
3. This range is helpful as it includes all 3 digit numbers that can be reversed to create a code. If a higher input were to be used, the program would only read the 3 right-most integer values. 1 or 2 digit inputs calculate an output with assumed leading 0's, meaning 5 becomes 005 and therefore FAA. In short, 0<x<100 are technically valid values.
4. If even, sum%2 is 0 and therefore there is no shift. If odd, sum % 2 is 1 and the shift is added.

```
char alpha = 'A' + ( (sum % 2) * ('a' - 'A') );
```

```
printf("Difference between letters\n");
printf(" 'a' - 'A' = %d\n", ('a' - 'A'));
printf(" 'j' - 'J' = %d\n", ('j' - 'J'));
printf(" 'r' - 'R' = %d\n", ('r' - 'R'));
printf(" 'z' - 'Z' = %d\n", ('z' - 'Z'));
```

5. The difference found should be 32. It is the same for all letters. Highlight than lowercase has higher ASCII value than uppercase.

```
// Round 4: Thousands Digit
rem = tmp % 10;
// printf ("%d\n" , rem );
tmp = tmp / 10;
acc = ( acc * 10) + rem ;

// Round 5: Ten Thousands Digit
rem = tmp % 10;
// printf ("%d\n" , rem );
tmp = tmp / 10;
acc = ( acc * 10) + rem ;
```

6.

```
// Implicitly Takes Care of the Left - Padding ( with type - casting )
char ch5 = ( char )( acc / 1 % 10) + alpha ; // Ones
char ch4 = ( char )( acc / 10 % 10) + alpha ; // Tens
char ch3 = ( char )( acc / 100 % 10) + alpha ; // Hundreds
char ch2 = ( char )( acc / 1000 % 10) + alpha ; // Thousands
char ch1 = ( char )( acc / 10000 % 10) + alpha ; // Ten-Thousands


// Print the Hidden Code
printf (" Hidden Code : %c%c%c%c%c\n", ch1 , ch2 , ch3, ch4, ch5 );
```

The computations for the sum must be updated, as well as the printing of the secret code. This is done by repeating the sum process twice more and by restructuring our secret code methodology.